



EÖTVÖS LORÁND UNIVERSITY

FACULTY OF INFORMATICS

DEPT. OF SOFTWARE TECHNOLOGY AND METHODOLOGY

ELTE Informatics Chatbot Using a Small Language Model

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Chapter 1

Introduction

1.1 Motivation

Students and staff at the Faculty of Informatics of Eötvös Loránd University face a recurring practical challenge: finding accurate, up-to-date answers to questions about the curriculum. Which courses must be completed before enrolling in a given subject? What are the rules for repeating a failed exam? How does the Neptun system handle course registration? The answers to these questions exist — scattered across faculty web pages, downloadable PDF regulations, DOCX course descriptions, and administrative guidelines — but locating the right document, navigating to the right section, and extracting a clear answer can take considerably longer than it should.

General-purpose search engines return ranked lists of pages; they do not synthesise an answer from multiple sources or understand the specific terminology of Hungarian higher education. As the volume of documentation grows each semester and more information migrates to sub-pages that are hard to discover by browsing, the gap between the information that exists and the information that students can efficiently find continues to widen.

1.2 Related Work

Several universities have built AI-powered platforms that demonstrate the viability of language models in higher education. Examining them critically, however,

reveals a consistent set of design assumptions that make them fundamentally different from the system developed in this thesis.

UM-GPT (University of Michigan) is a university-wide AI platform running on Microsoft Azure, offering access to multiple large language models to all students, faculty, and staff [1]. Its companion tool, Maizey, is built on RAG and allows any user to upload a custom dataset and create a personalised chatbot from it. While Maizey shares the same underlying retrieval mechanism as the system in this thesis, it is a general-purpose tool: users define their own knowledge base and their own purpose. The system developed here is purpose-built — its knowledge base is fixed to official faculty documents and its goal is to answer a specific and recurring student need, not to provide a blank canvas for arbitrary tasks.

ZotGPT (UC Irvine) is a broad AI services platform whose most relevant tool, ClassChat, allows instructors to configure an AI assistant around their course syllabus and materials [2]. Its primary orientation is pedagogical — supporting teaching and learning within a classroom context. It does not address the broader institutional information problem: the rules, procedures, and regulations that exist outside any single course and that students need to navigate independently.

CreateAI (Arizona State University) aggregates over 50 AI models into a single platform for building custom AI applications across the entire university [3]. Its defining characteristic is breadth, which is also its limitation for the problem at hand: a platform built to support dozens of unrelated domains simultaneously is not optimised for deep, precise retrieval over a single faculty’s regulatory documents.

Across all three systems, a common profile emerges: cloud infrastructure, general-purpose scope, and significant IT investment. None are designed to answer a specific question set drawn from a single document corpus, and none were evaluated on whether they actually answer domain-specific questions correctly. This is the design space the present work inhabits.

1.3 Problem Statement

The related systems above establish that LLM-powered question answering is viable in higher education. They do not, however, provide a usable template for the ELTE Faculty of Informatics. Two concrete problems block a direct adoption.

Domain knowledge and hallucination. General-purpose LLMs have no reliable knowledge of ELTE-specific regulations, course codes, or administrative procedures. Without grounding in official faculty documents, a language model will hallucinate plausible-sounding but incorrect answers — precisely the failure mode that is most harmful in an academic advising context, where a wrong answer about prerequisites or exam rules can have real consequences for a student’s academic progress.

Resource and infrastructure constraints. Deploying and maintaining a cloud-hosted platform at the scale of UM-GPT or CreateAI requires dedicated IT teams and substantial recurring costs. A single faculty cannot replicate that infrastructure. Any practical solution must run on commodity hardware — a standard laptop or a modest server — with no GPU and no cloud subscription.

No dedicated, locally deployed question-answering system currently exists for the ELTE Faculty of Informatics. Students and staff rely on direct web search, email queries to administrative offices, or informal peer advice — all of which are slow, inconsistently accurate, and do not scale as the faculty grows.

1.4 Objective

This thesis designs, implements, and evaluates a locally deployed question-answering system for the ELTE Faculty of Informatics. The central objective is to demonstrate that a system built on retrieval-augmented generation can provide students and staff with accurate, document-grounded answers to questions about the curriculum and administrative procedures — without requiring the infrastructure investment or general-purpose scope of the platforms examined in the related work. The specific goals of the system are:

- To collect and index publicly available faculty documents — including web pages, PDF regulations, and course descriptions — into a searchable local knowledge base that can be kept up to date as the faculty’s web presence changes.
- To answer natural language questions about the curriculum, course requirements, and administrative procedures by retrieving relevant document pas-

sages and generating grounded responses, rather than relying on the general knowledge of a language model.

- To run entirely on commodity hardware without a GPU or cloud subscription, demonstrating that an accurate and responsive question-answering system does not require significant infrastructure investment.
- To provide a browser-based chat interface with source references, so that users can verify the origin of each answer and follow up on the source documents directly.
- To evaluate the system empirically against a no-RAG baseline, measuring whether retrieval meaningfully improves answer accuracy on questions drawn from real student queries.
- To build the system on a web service architecture that is deployment-ready, so that moving from a local prototype to a live university service would require infrastructure provisioning only, with no changes to the application code itself.

1.5 Technologies Used

Two hard constraints shaped every technology decision: the system must run on commodity hardware with 8 GB of RAM and no discrete GPU, and no query or document may leave the local machine. The following components form the complete stack.

- **Ollama**, a local model server, packages and serves large language models over a stable REST interface, handling quantisation and memory mapping automatically. It abstracts the complexity of model loading so that the rest of the application calls a single endpoint regardless of which model is active. Switching models requires only a one-line change in the configuration file, with no modifications to application code.
- **Llama 3.2 (3B) and Gemma 3 (4B)**, the two language models evaluated in this thesis, are open-weight models small enough to run on consumer hardware. Both are served at 4-bit quantisation, which reduces their memory footprint to approximately 2–2.5 GB and makes CPU-only inference feasible. This keeps

the total memory usage well within the 8 GB RAM constraint while leaving headroom for the rest of the stack.

- **all-MiniLM-L6-v2**, a compact sentence transformer, encodes both document chunks and user queries into 384-dimensional dense vectors for semantic similarity search. Because the same model encodes both the index and the query, retrieval surfaces passages that are semantically close to the question rather than merely sharing keywords. It runs efficiently on CPU and provides a strong accuracy-to-resource trade-off for English factual retrieval tasks.
- **ChromaDB**, a lightweight vector database, stores the chunk embeddings and answers nearest-neighbour queries at runtime using cosine similarity. Unlike most vector databases, it runs in-process with no separate server and persists to a plain directory on disk. This satisfies the offline-operation requirement with no additional infrastructure to install or maintain.
- **FastAPI**, an asynchronous Python web framework, exposes the RAG pipeline as a REST API and serves the browser interface as static files from the same process. It was chosen for its low boilerplate, built-in request validation, and automatic API documentation. These properties make it practical for a single developer wrapping one core pipeline into a deployable HTTP service.
- **SQLite**, a serverless embedded database engine, logs every interaction — query, answer, sources, and response time — to a single file on disk with no server process required. It integrates directly into the Python application and requires no configuration beyond a file path. For a single-machine deployment with modest log volume, this is a simpler and more portable fit than a full database engine.
- **Vanilla JavaScript** powers the browser interface as a single self-contained HTML file with embedded CSS, served directly by the FastAPI backend. Avoiding a frontend framework eliminates a build step and all Node.js dependencies entirely. This keeps deployment as simple as placing one file in the static directory, consistent with the goal of minimal operational overhead.
- **Docker**, a containerisation platform, packages the entire application — the FastAPI backend, Ollama, and all dependencies — into isolated containers

that run identically regardless of the host environment. Each service is defined in a `docker-compose.yml` file, so the full stack starts with a single command and requires no manual dependency installation. This makes it the mechanism behind the deployability requirement: moving from a local machine to a faculty server requires only infrastructure provisioning, with no changes to the application code.

1.6 Thesis Structure

The remainder of this thesis is organised as follows.

Chapter 2 defines the system requirements. It states the functional requirements covering the capabilities exposed to the user, and the non-functional constraints including resource limits, latency, privacy, and offline operation.

Chapter 3 provides user documentation: a use-case diagram, step-by-step instructions for the web interface and command-line client, and annotated example queries.

Chapter 4 covers developer documentation. It describes the system architecture and component design, then details the full implementation across the data pipeline, embedding, RAG backend, REST API, logging, frontend, and deployment.

Chapter 5 presents the testing and evaluation of the system. It covers functional testing, the empirical RAG vs. baseline evaluation, and a user evaluation.

The Conclusion summarises the findings, reflects on the key design decisions, and outlines directions for future work.

Chapter 2

Software Requirements

This chapter defines what the ELTE IK Assistant must do and under what constraints it must operate. The functional requirements describe the capabilities the system exposes to the user. The non-functional requirements define the resource, privacy, latency, and reliability constraints that any correct implementation must satisfy. Interaction logging and the offline data pipeline are internal system concerns and do not appear as user-facing requirements.

2.1 Functional Requirements

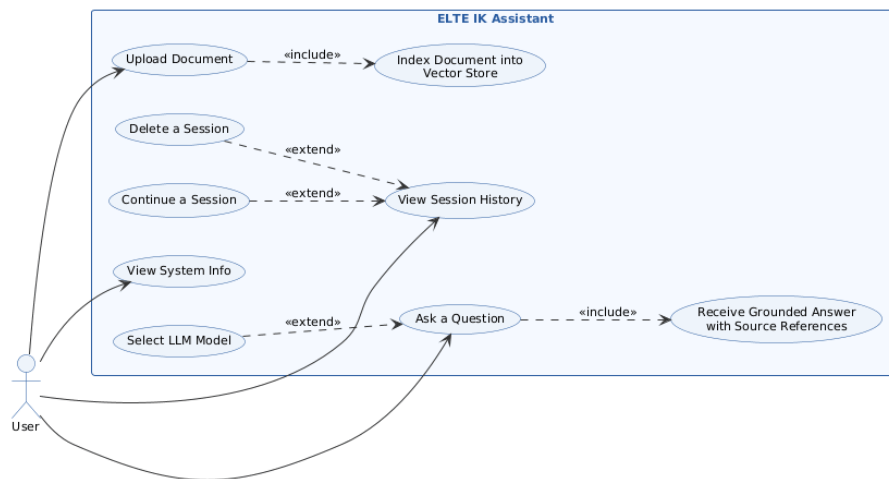


Figure 2.1: Use case diagram for the ELTE IK Assistant

Figure 2.1 illustrates the primary interactions between the user and the system. The single actor is the **User**, representing any student or staff member accessing the assistant through the browser interface. Asking a question is the central use case: it

always includes receiving a grounded answer with source references, and optionally extends to selecting a language model before submitting. Document upload always includes indexing the uploaded file into the vector store. Viewing session history optionally extends to continuing or deleting an existing session.

R-F-1 Natural language querying: the system accepts a free-text English question and returns a textual answer with source citations. When the answer is absent from the indexed documents, the system must say so explicitly rather than generating an unsupported response.

R-F-2 Model selection: the user may select which language model handles the current query. The system must apply the selected model to all subsequent queries in the session without requiring a restart.

R-F-3 Document upload: the user may upload a document that is immediately chunked, embedded, and added to the vector store, making its contents available for retrieval in the same session.

R-F-4 Embedding and retrieval: each chunk is encoded into a dense vector and stored in a local persistent vector database. At query time the top- K most similar chunks are retrieved; K is configurable.

R-F-5 Session history: the system persists conversation history across page reloads. The user may view past sessions, continue a previous session, or delete a session entirely.

R-F-6 System information: the user may view the current system status, including model availability and vector store state, without submitting a query.

R-F-7 REST API: the backend exposes a `/health` endpoint reporting system status and a `/chat` endpoint that accepts a message and returns the answer with source references. Both endpoints must be usable independently of the browser interface.

R-F-8 Web interface: a browser-based chat UI displays conversation history, source file names, model availability, and inline error messages without requiring any installation on the client side.

2.2 Non-Functional Requirements

R-NF-1 Privacy: no query, chunk, or answer may leave the local machine. The system must remain fully functional in a network-isolated environment after initial setup.

R-NF-2 Resource constraints: the system must run on 8 GB RAM with no discrete GPU, using a quantised (4-bit or 8-bit) language model and a CPU-friendly embedding model.

R-NF-3 Response latency: end-to-end response time must not exceed 90 seconds on the reference hardware. Embedding and retrieval must complete in under one second; generation typically takes 30–80 seconds on CPU.

R-NF-4 Offline operation: after the initial model pull and corpus crawl, no network connection is required. All subsequent pipeline steps run entirely from local storage.

R-NF-5 Reliability: if the language model process is unavailable, `/health` must report it and `/chat` must return an HTTP 503. The vector store and log database must not be corrupted on unexpected shutdown, and re-running the pipeline must be idempotent.

R-NF-6 Configurability: all tunable parameters — model names, K , temperature, paths, timeouts — are defined in a single configuration module. Swapping models requires only a one-line change there; changing the embedding model additionally requires rebuilding the vector store.

R-NF-7 Deployability: the system must be deployable on a faculty server without modification to the application code. All components are containerised with Docker, so that moving from a local prototype to a live university service requires infrastructure provisioning only — configuring ports, paths, and model names — with no changes to any other part of the codebase.

Chapter 3

User Documentation

This chapter describes how to set up and use the ELTE IK Assistant. Two setup paths are provided: a Docker-based path and a manual path. Choose based on your situation — Docker is better for isolated or shared deployments, the manual path is simpler for a single machine.

3.1 System Requirements

Table 3.1 lists the minimum system requirements for running the ELTE IK Assistant.

Component	Requirement
Operating system	Windows 10/11, macOS 12+, or Ubuntu 22.04+
RAM	8 GB minimum (16 GB recommended)
Disk space	$\approx$ 6 GB (models + vector store + crawled data)
Internet access	Required during first-run setup only

Table 3.1: Minimum system requirements.

3.2 Installation and Configuration

Option A — Docker

Docker Compose orchestrates all services automatically: the Ollama language model server, both model downloads, the knowledge base build, and the FastAPI application. This option is best suited for isolated or shared deployments.

First-run note: on the first `docker compose up`, the container crawls ELTE documents and builds the vector index before starting the server. This takes approximately 5–10 minutes depending on internet speed and hardware. Subsequent starts are immediate.

1. Install **Docker Desktop** (includes Docker Compose) from the official Docker website for your operating system.
2. Clone or download the project repository:

```
1 git clone https://github.com/borohull/Thesis.git
2 cd Thesis/elte_chat
```

3. Start all services:

```
1 docker compose up -d
```

4. Monitor first-run progress:

```
1 docker compose logs -f app
```

Wait until `Application startup complete` appears in the logs.

5. Open the chat interface at `http://localhost:8000/static/index.html`.

6. To stop the containers:

```
1 docker compose down
```

Option B — Manual Setup

This option is best suited for running the assistant on a single local machine. Steps 1–4 are performed **once** by whoever sets up the system. End users only need step 5 (opening the browser).

Prerequisite: Python 3.10 or newer must be installed on your system.

1. Install **Ollama** from `https://ollama.com/download`, then start the service and pull both language models:

```
1 ollama serve
2 ollama pull llama3.2:3b
3 ollama pull gemma3:4b
```

2. Clone or download the project repository:

```
1 git clone https://github.com/borohull/Thesis.git
2 cd Thesis/elte_chat
```

3. Run the setup script. This creates a virtual environment, installs dependencies, crawls ELTE documents, and builds the vector index.

macOS / Linux:

```
1 chmod +x setup.sh
2 ./setup.sh
```

Windows:

```
1 setup.bat
```

The script prints progress at each stage. A complete first run takes approximately 5–10 minutes. Listing 3.2 shows the expected terminal output.

```
1 === ELTE IK Assistant Setup ===
2 [1/6] Starting Ollama service...
3 [2/6] Pulling language models (this may take several minutes)
4 ...
5 pulling manifest
6 pulling dde5aa3fc5ff: 100% 2.0 GB
7 ...
8 success
9 pulling manifest
10 pulling aeda25e63ebd: 100% 3.3 GB
11 ...
12 success
13 [3/6] Setting up Python environment...
14 [4/6] Crawling ELTE documents (this will take several minutes)
15 ...
16 [SAVE] https://inf.elte.hu/en -> index.html
17 [SAVE] https://inf.elte.hu/en/courses -> courses.html
18 [PDF-SAVE] https://inf.elte.hu/... -> regulations.pdf
19 [SAVE] https://www.elte.hu/en/quaestura-office -> quaestura
20 -office.html
21 ...
22 Crawl complete -- saved: 482, skipped: 0, failed: 5, PDFs: 754,
23 DOCXs: 46
```

```
20 [5/6] Building knowledge base (chunking + embedding)...
21 === build_index.py: ELTE IK knowledge base builder ===
22 --- Phase 1: Ingestion ---
23 Scan: 482 files on disk, 0 in manifest
24   new:          482
25   updated:      0
26   deleted:      0
27   unchanged:    0
28 --- Phase 2: Embedding ---
29 Loaded 6128 chunks
30 Batches: 100% 192/192
31 Upserted 6128 chunks into 'elte_ik'
32 Collection now has 6128 documents
33 === Done. Knowledge base is up to date. ===
34 [6/6] Setup complete.
35
36 To start the server:
37   source .venv/bin/activate && uvicorn app.main:app  (macOS/
      Linux)
38   .venv\Scripts\activate && uvicorn app.main:app      (Windows)
39
40 Then open: http://localhost:8000/static/index.html
```

Code 3.1: Expected terminal output during first-run setup.

On subsequent runs the crawler skips already-downloaded files and the embedding step skips unchanged chunks, so re-running setup is fast and safe.

4. Start the server:

```
1 # macOS / Linux
2 source .venv/bin/activate && uvicorn app.main:app
3
4 # Windows
5 .venv\Scripts\activate && uvicorn app.main:app
```

Code 3.2: Expected terminal output during first-run setup.

5. Open the chat interface at `http://localhost:8000/static/index.html`.

To verify the server is running correctly, run:

```
1 curl http://localhost:8000/health
```


A successful response is `{"status": "ok", "ollama": true}`. If `"ollama": false` appears, Ollama is not running or the selected model has not been pulled.

Configuration

All tunable parameters are defined in `app/settings.py`. The most commonly adjusted values are listed in Table 3.2.

Parameter	Default	Description
OLLAMA_MODEL	llama3.2:3b	Language model served by Ollama
EMBEDDING_MODEL	all-MiniLM-L6-v2	Sentence embedding model
TOP_K	3	Chunks retrieved per query
TEMPERATURE	0.1	LLM sampling temperature
CHROMA_PATH	data/processed/chroma_store	Chroma store location on disk

Table 3.2: Key configuration parameters in `app/settings.py`.

Changing `OLLAMA_MODEL` takes effect on the next server start. Changing `EMBEDDING_MODEL` additionally requires rebuilding the vector store by re-running `python scripts/build_index.py`.

Note: these settings are relevant only for whoever deploys the system, not for end users asking questions through the interface.

3.3 Using the Web Interface

Open `http://localhost:8000/static/index.html` in any modern browser once the server is running. The interface consists of a left sidebar listing chat sessions, a central chat panel, and a right sidebar for model selection. A status indicator in the header shows whether Ollama is reachable — if it shows as unavailable, start or restart Ollama before sending a query.

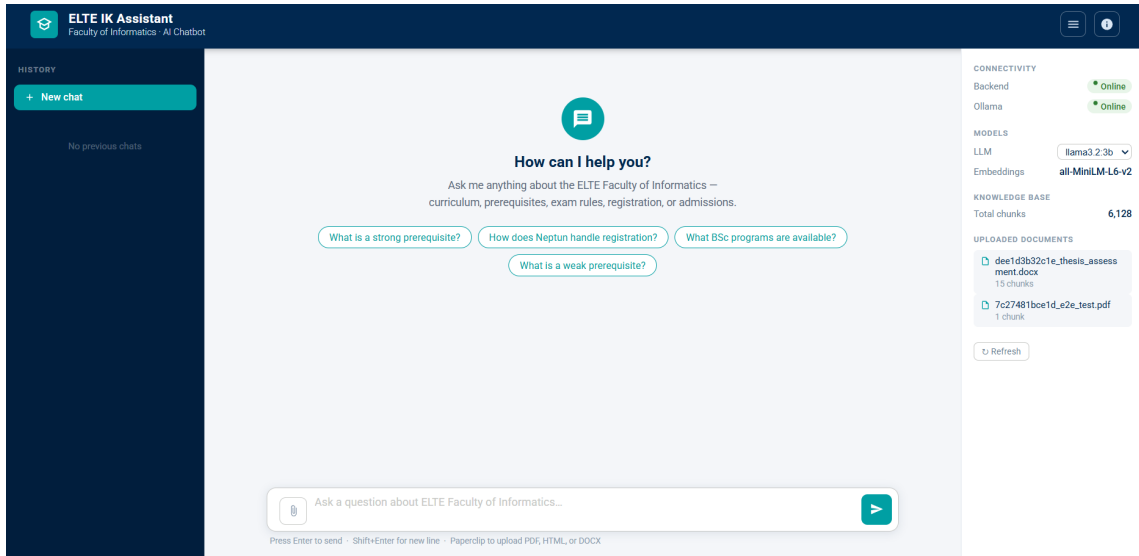


Figure 3.1: The ELTE IK Assistant web interface.

Asking a question. Type a question in the input field at the bottom of the chat panel and press **Enter** or click the send button. The assistant retrieves relevant document chunks, constructs a grounded prompt, and returns an answer along with the source file names it used. Questions should relate to the ELTE Faculty of Informatics — curriculum, regulations, procedures, and student services. On a CPU-only machine with 8 GB of RAM, typical response times are 30–80 seconds per query. The first query after startup may be slower while the model loads into memory.

Source references. Every answer displays the names of the source documents used to construct it (for example `elte_ik_regulations.html` or `TAJ.pdf`). These are the actual files retrieved from the knowledge base. If an answer looks incorrect or incomplete, the source names indicate which documents to check directly.

Selecting a language model. The right sidebar lists all Ollama models available on the machine. Click a model name to switch for the current session. `llama3.2:3b` is the default and works on most hardware. `gemma3:4b` is also available and may produce different quality results.

Managing chat sessions. Each conversation is stored as a named session. Sessions appear in the left sidebar and persist across browser refreshes and server restarts. Click **New chat** to start a new conversation, or hover over an existing session and click the $\times$ button to delete it.

Uploading custom documents. The knowledge base can be extended with your own files. Click **Upload document** above the input field, select a PDF, HTML, or DOCX file (maximum 20 MB), and wait for the confirmation message. The document is chunked, embedded, and available for retrieval immediately — no server restart required.

3.4 Using the Command-Line Client

A lightweight CLI client is available for scripting and testing without a browser. The FastAPI server must be running before starting the client:

```
1 python scripts/chat_cli.py
```

Type a question at the **You:** prompt and press Enter. The client sends the question to the server and prints the answer and source references. Type **exit** or press **Ctrl+C** to quit.

3.5 Example Queries

The following examples illustrate the range of questions the system handles and how it responds to out-of-scope input.

Example 1 — Prerequisite query. *Question:* “What is a strong prerequisite and how does it differ from a weak one?”

Answer: According to the context, a strong prerequisite is a prerequisite without which the follow-up subject cannot be taken in the next semester, and the follow-up subject can only be registered for after the prerequisite subject has been fulfilled. It must be completed before taking the follow-up subject. A strong prerequisite differs from a weak one in that it has no special annotation (as opposed to being marked as 'weak') and is considered essential for taking the follow-up subject in the next semester.

Sources: prerequisites.html

Example 2 — Exam rules. *Question:* “How many times can I retake a failed exam in one semester?”

Answer: According to Section 79 (6) of the ELTE Academic Regulations, you can retake and improve your grade in no more than two exams per semester.

Sources: ELTE Academic Regulations.pdf

Example 3 — Ranking when registering for a subject. *Question:* “How do I see my ranking when registering for a subject?”

Answer: To see your ranking when registering for a subject, follow these steps:

1. Find the subject you want to register for. 2. Click the arrow icon at the end of the corresponding row. 3. Then click "Details" to view your score and your position in the ranking. This will show you your ranking sequence and lucky point (ranking score 1 and 2, respectively)

Sources: registration for subjects BSc_MSc (1).pdf

Example 4 — Out-of-scope query. *Question:* “What is the capital city of Australia?”

Answer: The information needed to answer this question is not available in the documents I have access to. I can only answer questions related to the ELTE Faculty of Informatics curriculum, regulations, and procedures.

Sources: (none)

The fourth example demonstrates the system’s refusal behaviour: when no relevant chunks are retrieved, the assistant explicitly acknowledges the gap rather than generating an unsupported answer.

Chapter 4

Developer Documentation

This chapter covers the full developer documentation for the ELTE IK Assistant. It begins with the system architecture and key design decisions, then details the implementation of each layer — from data collection and embedding through the RAG backend, REST API, and frontend — and concludes with setup and deployment instructions.

4.1 System Architecture

The ELTE IK Assistant is organised as a three-layer pipeline:

1. **Offline data pipeline** — crawls ELTE websites, cleans and chunks the raw documents, encodes the chunks into vector embeddings, and stores them in a local vector database. This pipeline is run once by the system administrator and re-run whenever the source documents change.
2. **Online serving layer** — a FastAPI web server that accepts a user’s natural-language question, retrieves the most relevant document chunks from the vector database, builds a prompt, calls a locally running language model, and returns the generated answer together with source references.
3. **Presentation layer** — a single-page browser UI that maintains session state, renders the conversation, and displays source file metadata alongside each answer.

Figure 4.1 illustrates the high-level data flow between these layers.

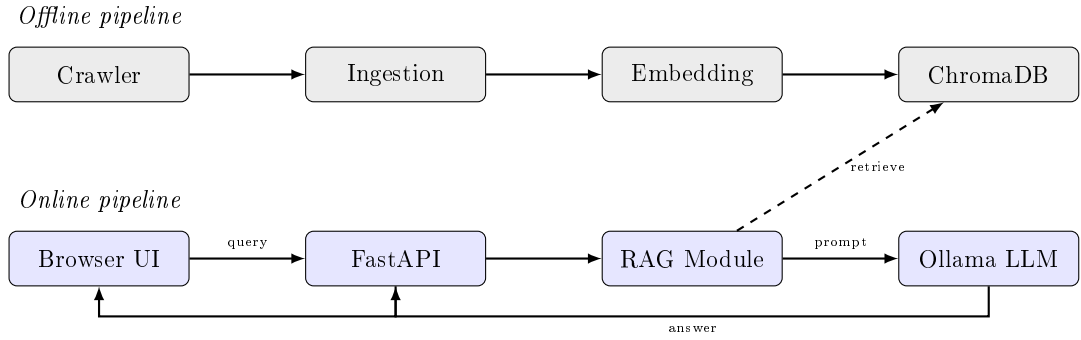


Figure 4.1: High-level architecture of the ELTE IK Assistant. The offline pipeline (top) is run once to build the vector index; the online pipeline (bottom) handles every user query at runtime.

The strict separation between the offline and online pipelines is a deliberate consequence of the offline-operation requirement (R-NF-1): once the vector index is built, the serving layer never reads from disk beyond the ChromaDB query and needs no network access.

4.2 Technology Choice

The system was developed and tested on a personal laptop running Windows 11 with 8 GB of RAM and no dedicated GPU. Beyond the hardware, a core requirement was that no user query or document ever leave the local machine, which ruled out cloud-hosted language model APIs. These two constraints — limited memory and strict data locality — drove every component choice toward tools that are lightweight, run in-process, and require no external services.

Concern	Choice	Reason
Language model serving	Ollama	Preferred over running a raw <code>llama.cpp</code> binary because it exposes a stable REST API, handles model quantisation and memory mapping automatically, and makes switching models a one-line change in <code>settings.py</code> .
Language model	Llama 3.2 (3B) / Gemma 3 (4B)	The 8 GB RAM constraint rules out larger models at 16-bit precision. Both models use 4-bit quantisation, reducing their footprint to approximately 2–2.5 GB and leaving sufficient headroom for the OS, embedding model, and vector store.
Embeddings	all-MiniLM-L6-v2	A compact sentence transformer that produces 384-dimensional vectors and runs efficiently on CPU without a dedicated GPU. Although larger embedding models exist, this model offers a strong accuracy-to-resource trade-off for English factual retrieval tasks.
Vector store	ChromaDB	Runs in-process with no separate server, persists to a plain directory, and supports cosine-similarity search out of the box. This satisfies the offline-operation requirement with minimal operational overhead.
Backend framework	FastAPI	Asynchronous Python framework with automatic OpenAPI documentation, built-in static file serving, and low boilerplate. Well suited to wrapping a single RAG pipeline as a REST API.
Interaction log	SQLite	The application runs on a single machine and log volume is modest. SQLite requires no separate server process and stores the entire database in a single portable file.
Frontend	Vanilla JS	A framework-free single HTML file eliminates a build step and all Node.js dependencies, keeping deployment as simple as copying a file into <code>app/static/</code> .

Table 4.1: Technology stack and rationale for each choice.

4.3 Component Design

Figure 4.2 shows the module structure of the application and the dependencies between them.

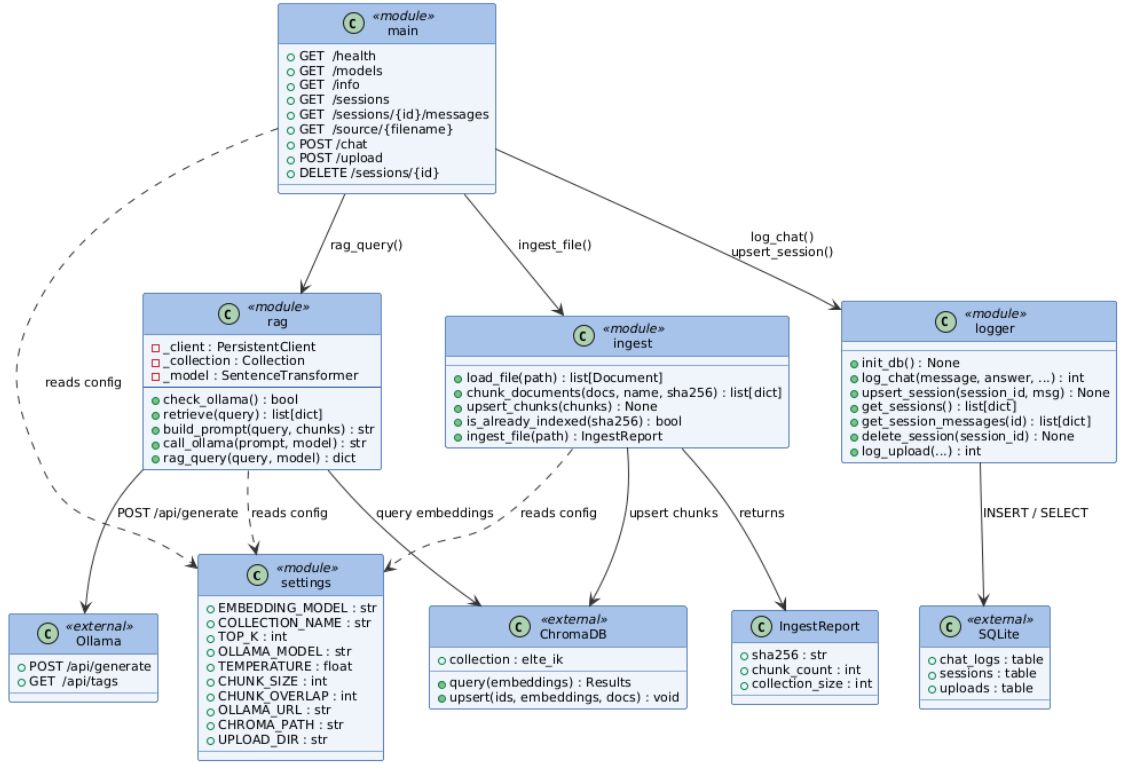


Figure 4.2: Class diagram of the ELTE IK Assistant application modules.

- Crawler:** performs a breadth-first traversal of the ELTE Faculty of Informatics website starting from a set of seed URLs. It downloads HTML pages, PDF documents, and DOCX files into `data/raw/`, computes a SHA-256 digest of each file, and records it in a manifest. On subsequent runs only files whose digest has changed are re-downloaded, keeping re-indexing time proportional to the volume of changed content. It feeds directly into the Ingestion component.
- Ingestion and Chunking:** receives raw files from the Crawler and applies format-specific loaders (PyMuPDF for PDFs, BeautifulSoup for HTML, python-docx for DOCX) to extract clean text. The text is split into overlapping 800-character chunks with a 150-character overlap using a recursive character splitter. Each chunk carries metadata (source path, file type, page number, chunk identifier) and is written to `data/processed/chunks.json` for the Embedding component.
- Embedding Model and Vector Store:** the all-MiniLM-L6-v2 sentence-transformer [4] encodes each chunk into a 384-dimensional vector. The model occupies approximately 90 MB and runs entirely on CPU. Vectors are stored

in a ChromaDB collection named `elte_ik` using cosine similarity, persisted to `data/processed/chroma_db/`. The Vector Store is the boundary between the offline pipeline and the online serving layer.

- **RAG Module:** the core of the online pipeline. At query time it encodes the user’s question with the same `all-MiniLM-L6-v2` model (mandatory — a model mismatch produces wrong retrieval results), retrieves the top- k most similar chunks from ChromaDB by cosine similarity, assembles them into a grounded prompt that instructs the LLM to answer only from the provided context, and forwards the prompt to the Language Model. Both the ChromaDB client and the embedding model are initialised once at server startup as singletons, eliminating per-request load overhead.
- **FastAPI Backend:** exposes the RAG pipeline over HTTP. The `POST /chat` endpoint calls the RAG Module and logs the result; `GET /health` reports whether Ollama is reachable; additional endpoints handle session management and user-uploaded documents. Static files (the single-page UI) are served from `app/static/` via FastAPI’s `StaticFiles` mount.
- **Chat Logging:** every request–response pair is persisted to a SQLite database (`data/logs/chat_logs.db`). The schema records the session identifier, user message, assistant answer, source file list, response time, and timestamp. Logging supports both retrospective quality analysis and the session sidebar in the UI.
- **Frontend:** a single self-contained HTML file with embedded CSS and vanilla JavaScript served by the FastAPI backend. It communicates with the backend exclusively via REST and holds no local copy of the vector index or the language model. The framework-free design eliminates a build step and reduces operational complexity.

4.4 Data Flow

Figure 4.3 shows the complete data flow for a single user query. The main path runs top to bottom through the serving layer; ChromaDB and the SQLite log are accessed as side effects of the central processing steps.

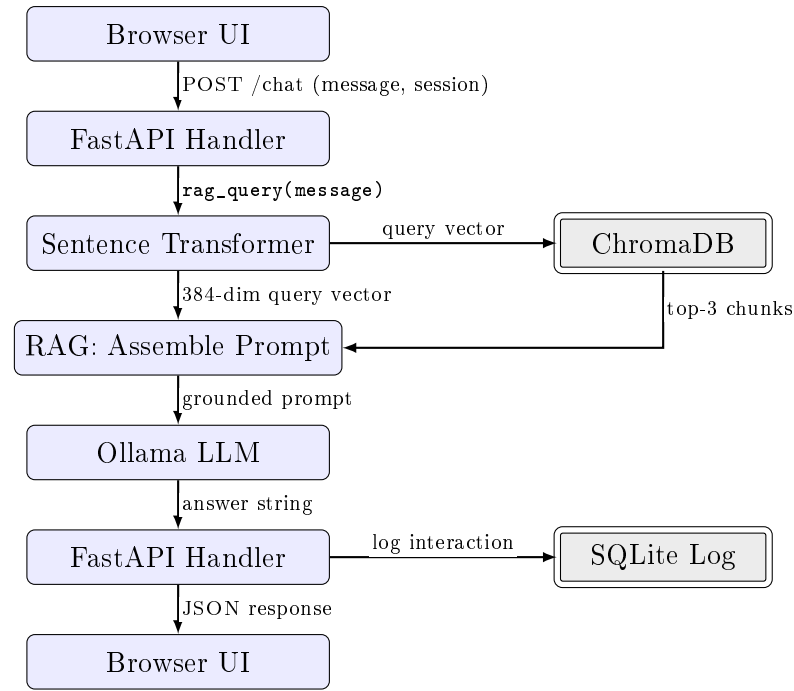


Figure 4.3: Data flow for a single user query. The main path (left column) shows the request processing sequence; ChromaDB is queried during prompt assembly and the interaction is logged to SQLite before the response is returned.

The round-trip latency is dominated by the Ollama inference step. On a CPU-only machine with `llama3.2:3b` the median observed latency is approximately 48 seconds without RAG context and 70 seconds with RAG context. With `gemma3:4b` the figures are 121 seconds and 61 seconds respectively — the shorter RAG latency reflects the model generating a more focused answer when constrained to the provided context.

4.5 Storage Design

The system uses two persistent stores, each chosen to match the access pattern of the data it holds.

Store	Location	Contents and access pattern
ChromaDB	<code>data/processed/chroma_db/</code>	6112 chunk embeddings (384-dim float32) + metadata. Written once by the offline pipeline; read on every query via approximate nearest-neighbour search.
SQLite	<code>data/logs/chat_logs.db</code>	One row per chat turn: session id, user message, assistant answer, sources, timestamp. Appended on every request; queried for session listing and history.
Manifest	<code>data/processed/manifest.json</code>	SHA-256 digests of all crawled files. Read and written by the crawler to detect changed files during incremental re-indexing.

Table 4.2: Persistent storage used by the ELTE IK Assistant.

All three stores are plain files on the local filesystem. No database server, no cloud storage, and no network filesystem are involved, satisfying the offline-operation and data-privacy requirements.

4.6 Key Design Decisions

Local LLM via Ollama. Running the language model locally is the single most consequential architectural decision. It satisfies the privacy requirement (no query leaves the machine) and the offline requirement (no internet connection needed at inference time), at the cost of significantly higher latency compared with a cloud API. Ollama was chosen over running a raw `llama.cpp` binary because it exposes a stable REST interface, handles model quantisation and memory mapping automatically, and supports multiple models without code changes — the model name in `settings.py` is the only thing that changes when switching between `llama3.2:3b` and `gemma3:4b`.

Small quantised models. The 8 GB RAM constraint rules out 7B+ parameter models at 16-bit precision (≈ 14 GB). Both evaluated models (`llama3.2:3b` and `gemma3:4b`) use 4-bit quantisation, which reduces their memory footprint to approximately 2 GB and 2.5 GB respectively, leaving enough headroom for the embedding model, ChromaDB index, and operating system.

Top- $k = 3$ retrieval. Three chunks were chosen as the default retrieval count after balancing two competing forces: more chunks provide richer context but increase the prompt length, which in turn increases inference latency and can cause the model to lose focus on the most relevant passage. At 800 characters per chunk, three chunks contribute approximately 2400 characters of context — well within the 4096-token context window of the evaluated models while keeping the prompt concise.

Temperature = 0.1. A near-zero temperature makes the language model near-deterministic and biased toward its highest-probability completions. For a factual question-answering chatbot this is desirable: users expect consistent, precise answers rather than creative variation. The small non-zero value (rather than exactly 0) avoids degenerate repetition loops that some models exhibit at temperature zero.

Separation of offline and online pipelines. The embedding pipeline is entirely offline and produces an artefact (the ChromaDB directory) that the serving layer consumes read-only. This means the serving layer starts instantly — there is no warm-up embedding step — and the administrator can rebuild the index without touching the running server. It also means the embedding model and the vector store are initialised once at server startup rather than on every request, which is essential for meeting the latency target.

4.7 Implementation

This section describes the full implementation of the ELTE IK Assistant across its four layers: data collection and ingestion, document embedding and vector storage, the RAG backend, and the web-based frontend. Each subsection covers one layer in detail, including the key design choices, data structures, and code.

4.7.1 Setup and Deployment

Table 4.3 lists the minimum system requirements.

Component	Requirement
Operating system	Windows 10/11, macOS 12+, or Ubuntu 22.04+
RAM	8 GB (16 GB recommended)
Disk space	≈6 GB (models + vector store + crawled data)
Python	3.10 or newer
Ollama	Latest release (https://ollama.com)
Internet access	Required only during the initial setup phase

Table 4.3: Minimum system requirements.

Install Ollama. Download and install Ollama from <https://ollama.com/download>, then start the background service and pull the two language models:

```
1 ollama serve
2 ollama pull llama3.2:3b
3 ollama pull gemma3:4b
```

Clone the repository and create a virtual environment.

```
1 git clone https://github.com/borohull/Thesis.git
2 cd Thesis/elte_chat
3 python -m venv .venv
4 # Windows: .venv\Scripts\activate
5 # macOS/Linux: source .venv/bin/activate
6 pip install -r requirements.txt
```

Build the knowledge base. This is a one-time process that must be completed before the server can answer questions.

1. Run the crawler to download ELTE documents into `data/raw/`:

```
1 python scripts/crawler.py
```

2. Open JupyterLab and run `notebooks/01_data_ingestion.ipynb` — loads and chunks the raw files, writes `data/processed/chunks.json`.
3. Run `notebooks/02_embedding.ipynb` — encodes chunks with `all-MiniLM-L6-v2` and stores vectors in ChromaDB under `data/processed/chroma_db/`.

All tunable parameters are defined in `app/settings.py`. The most commonly adjusted settings are shown in Table 4.4.

Setting	Default	Description
OLLAMA_MODEL	llama3.2:3b	Default language model
TOP_K	3	Chunks retrieved per query
TEMPERATURE	0.1	LLM sampling temperature
EMBEDDING_MODEL	all-MiniLM-L6-v2	Must match notebook 02

Table 4.4: Key configuration parameters in `app/settings.py`.

Important: `EMBEDDING_MODEL` must be kept consistent between the index build and the serving layer. If it is changed, notebook 02 must be re-run to rebuild the vector store — a mismatch between the query encoder and the stored index produces incorrect retrieval results.

Start the server. Run the following command from the project root with the virtual environment active and Ollama running:

```
1 uvicorn app.main:app --reload
```

The server starts on `http://localhost:8000`. Verify it is healthy with:

```
1 curl http://localhost:8000/health
```

A successful response is `{"status": "ok", "ollama": true}`. If `"ollama": false` appears, Ollama is not running or the selected model has not been pulled.

4.7.2 Central Configuration

All tunable parameters are centralised in a single settings module so that they can be changed in one place without touching any business logic.

```
1 from pathlib import Path
2
3 BASE_DIR = Path(__file__).parent.parent
4 CHROMA_PATH = str(BASE_DIR / "data/processed/chroma_db")
5 CHUNKS_PATH = str(BASE_DIR / "data/processed/chunks.json")
6 MANIFEST_PATH = str(BASE_DIR / "data/processed/manifest.json")
7
8 PENDING_CHANGES_PATH = str(BASE_DIR / "data/processed/
9     pending_changes.json")
10
11 RAW_DIR = str(BASE_DIR / "data/raw")
12
13 COLLECTION_NAME = "elte_ik"
14 EMBEDDING_MODEL = "all-MiniLM-L6-v2"
15 TOP_K = 3
```

```
12
13 OLLAMA_URL      = "http://localhost:11434/api/generate"
14 OLLAMA_MODEL    = "llama3.2:3b"
15 TEMPERATURE    = 0.1
16 TIMEOUT_S      = 120
```

Code 4.1: Central configuration (`app/settings.py`)

4.7.3 Data Collection and Ingestion

Before any machine learning component can be applied, the raw source material must be collected and converted into a uniform text format. ELTE Faculty of Informatics publishes its relevant information across several channels: static HTML pages, downloadable PDF documents (study regulations, course descriptions, exam rules), and plain-text files. The ingestion pipeline handles all three formats through a unified loader interface.

File Loaders

Each file type requires a different parsing strategy. PDFs are loaded using PyMuPDF (`pymupdf`), which extracts text page by page with reliable layout handling. HTML files are parsed with `BeautifulSoup`: structural noise tags (`script`, `style`, `nav`, `footer`, `header`, `aside`) are removed first, then content is extracted from the `<main>` or `<article>` element if present, with each paragraph prefixed by its nearest heading to preserve section context. DOCX files are parsed with `python-docx`, which extracts paragraphs and tables while tracking heading structure. Plain text and Markdown files are read directly. In every case the resulting text goes through a shared cleaning function that collapses runs of blank lines and normalises whitespace.

```
1 def clean_text(text: str) -> str:
2     text = re.sub(r"\n{3,}", "\n\n", text)
3     text = re.sub(r"[\t]+", " ", text)
4     return text.strip()
5
6 def load_file(file_path: str | Path) -> List[Document]:
7     file_path = Path(file_path)
8     suffix = file_path.suffix.lower()
```

```
9     if suffix == ".pdf":
10         return load_pdf_pymupdf(file_path)
11     elif suffix in {".txt", ".md"}:
12         return load_txt(file_path)
13     elif suffix in {".html", ".htm"}:
14         return load_html(file_path)
15     elif suffix == ".docx":
16         return load_docx(file_path)
17     else:
18         raise ValueError(f"Unsupported file type: {suffix}")
```

Code 4.2: Text cleaning and unified file loader (`01_data_ingestion.ipynb`)

Every document is represented as a `LangChain Document` object carrying both its text content and a metadata dictionary that records the original file name, file type, page number (for PDFs), and a `source_relative` path used by the incremental update pipeline. This metadata is preserved all the way into the vector database and is later surfaced to the user as source references.

Web Crawler

Raw documents are collected by a standalone crawler (`scripts/crawler.py`) that performs a resumable breadth-first crawl across two domains. It crawls all English pages under `inf.elte.hu` recursively, and twenty whitelisted path subtrees on `www.elte.hu` covering topics such as Neptun registration, visa procedures, student finances, and dormitory housing. Staying within explicit path prefixes on the second domain prevents the crawler from wandering into unrelated university-wide content.

For each HTML page the crawler checks the `lang` attribute and the URL path to confirm the page is in English before saving it, discarding Hungarian-language variants. Linked PDF and DOCX files discovered on any crawled page are downloaded separately into `data/raw/linked_pdfs/` and `data/raw/linked_docx/` respectively. A 0.5-second delay between requests limits server load.

The crawler is resumable: if a destination file already exists it is skipped, so an interrupted run can be continued without re-downloading pages that were already saved. A `-reset` flag wipes `data/raw/` and `data/processed/` and starts fresh. The final corpus contains 482 files spanning HTML pages, PDFs, and DOCX documents.

Breadth-First Search Traversal

The crawler traverses the web graph using Breadth-First Search (BFS), implemented with a `deque` as the frontier queue. BFS is the appropriate choice here because it explores pages level by level — the seed URLs are processed first, then pages one hop away, then two hops away, and so on. This guarantees that shallower (and therefore typically more important) pages are crawled before deep or obscure sub-pages, and it bounds memory usage to the width of the graph rather than its depth.

Algorithm 1 BFS Web Crawler

Require: Seeds S , allowed-domain predicate $allowed(\cdot)$

```
1:  $visited \leftarrow \emptyset$ 
2:  $queue \leftarrow \text{DEQUEUE}(S)$ 
3: while  $queue$  is not empty do
4:    $u \leftarrow queue.popleft()$ 
5:   if  $u \in visited$  then continue
6:   end if
7:    $visited += \{u\}$ 
8:   if  $u$  already on disk then parse cached HTML; goto link extraction
9:   end if
10:   $html \leftarrow \text{FETCH}(u)$ 
11:  if  $html$  is English then save to data/raw/
12:  end if
13:  for each link  $v$  extracted from  $html$  do
14:    if  $v$  ends in .pdf or .docx and  $allowed(v)$  then
15:      download  $v$  to linked_pdfs/ or linked_docx/
16:    else if  $allowed(v)$  and  $v \notin visited$  then
17:       $queue.append(v)$ 
18:    end if
19:  end for
20: end while
```

Chunking Strategy

Large documents cannot be stored or retrieved as single units: a language model has a limited context window, and retrieving a full 50-page PDF in response to a one-sentence question would waste context and degrade answer quality. The documents are therefore split into overlapping chunks using `RecursiveCharacterTextSplitter` from `LangChain`.

```
1 text_splitter = RecursiveCharacterTextSplitter(
2     chunk_size=800,
```

```
3     chunk_overlap=150,
4     length_function=len,
5     separators=["\n\n", "\n", ". ", " ", ""]
6 )
```

Code 4.3: Chunking configuration (01_data_ingestion.ipynb)

A chunk size of 800 characters was chosen as a practical balance: large enough to contain a complete thought or regulation clause, yet small enough for the embedding model and the LLM to process efficiently. The 150-character overlap ensures that information near a chunk boundary is not silently dropped. If a sentence is split between two chunks, both chunks contain it. The separator list defines a priority order: the splitter tries to break on double newlines first (paragraph boundaries), then single newlines, then sentence ends, and only falls back to splitting mid-word if no other boundary is available.

Chunks shorter than 100 characters (for example lone headers or page numbers) are discarded, and each surviving chunk is assigned a `chunk_id` of the form `<source_relative>::chunk_N` before being saved to a JSON file for the next stage.

Incremental Ingestion with SHA-256 Manifest

Re-processing all 482 source files from scratch every time a document is updated would be wasteful. The pipeline therefore maintains a *manifest* — a JSON file at `data/processed/manifest.json` — that records the SHA-256 content hash, chunk count, and last-processed timestamp for every file that has been ingested.

On each run, notebook `01_data_ingestion.ipynb` computes the SHA-256 hash of every file currently on disk and compares it against the manifest. This produces three change sets: new files (present on disk but not in the manifest), updated files (hash has changed), and deleted files (in the manifest but no longer on disk). Only new and updated files are re-chunked; unchanged files are skipped entirely.

```
1 new_files      = sorted(disk_paths - manifest_paths)
2 deleted_files = sorted(manifest_paths - disk_paths)
3 updated_files = sorted(
4     p for p in (disk_paths & manifest_paths)
5     if manifest[p]["content_hash"] != current_files[p][1]
6 )
```

Code 4.4: Incremental diff logic (01_data_ingestion.ipynb)

After chunking, the results of the diff are saved to `data/processed/pending_changes.json`. Notebook `02_embedding.ipynb` reads this file to know exactly which chunk IDs to remove from ChromaDB and which new embeddings to insert, avoiding a full re-embed of the collection. Once notebook 02 completes it deletes the pending changes file so that the next run starts clean.

4.7.4 Embedding and Vector Storage

With the text broken into chunks, the next step is to convert each chunk into a dense numeric vector (an *embedding*) that captures its semantic meaning. Chunks with similar meaning will have embeddings that are close to each other in vector space, which allows the retrieval step to find relevant content even when the user's phrasing differs from the exact wording in the source document.

Embedding Model

The chosen model is `all-MiniLM-L6-v2`, a compact sentence transformer from the Sentence-Transformers library that produces 384-dimensional vectors.

```
1 class EmbeddingPipeline:
2     def __init__(self, model_name: str = EMBEDDING_MODEL):
3         self.model = SentenceTransformer(model_name)
4
5     def encode(self, texts: list[str]) -> np.ndarray:
6         return self.model.encode(texts, show_progress_bar=True)
```

Code 4.5: Embedding pipeline class (`02_embedding.ipynb`)

All chunk texts are encoded in a single batched call, producing a matrix of shape `(num_chunks, 384)`. The same model instance is reused at query time inside the production application to ensure that document and query embeddings live in the same vector space.

ChromaDB Vector Store

The embeddings, original texts, and metadata are stored in ChromaDB, an open-source vector database that persists data to disk and supports fast approximate nearest-neighbour search. A single persistent collection named `elte_ik` is created

with cosine similarity as the distance metric, which is standard for sentence embeddings.

```
1 client = chromadb.PersistentClient(path=CHROMA_PATH)
2 collection = client.get_or_create_collection(
3     name=COLLECTION_NAME,
4     metadata={"hnsw:space": "cosine"}
5 )
6
7 collection.upsert(
8     ids=[str(c["metadata"]["chunk_id"]) for c in chunks],
9     embeddings=embeddings.tolist(),
10    documents=texts,
11    metadatas=[c["metadata"] for c in chunks]
12 )
```

Code 4.6: Upserting embeddings into ChromaDB (02_embedding.ipynb)

Using `upsert` rather than `add` means that re-running the embedding notebook after new documents have been scraped will update existing chunks in place rather than creating duplicates. The collection is stored under `data/processed/chroma_db/` and is read at runtime by the production RAG module without any re-processing.

4.7.5 Retrieval-Augmented Generation Pipeline

The RAG pipeline is the core intelligence of the system. It consists of three steps that run on every incoming user query: retrieval, prompt construction, and LLM generation.

Retrieval

When a user submits a question, the embedding model encodes it into a 384-dimensional query vector. ChromaDB then returns the `TOP_K` (default: 3) chunks whose stored embeddings are most similar to the query vector under cosine similarity.

Cosine Similarity

Cosine similarity measures the angle between two vectors rather than their Euclidean distance. For a query embedding $\mathbf{q}$ and a chunk embedding $\mathbf{d}$, it is defined

as:

$$\text{sim}(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \|\mathbf{d}\|} \quad (4.1)$$

The result lies in $[-1, 1]$; a value of 1 means the vectors point in exactly the same direction (maximally similar), and 0 means they are orthogonal (unrelated). Cosine similarity is preferred over Euclidean distance for sentence embeddings because it is invariant to the magnitude of the vectors — two chunks that express the same concept but with different amounts of text will score similarly, whereas Euclidean distance would penalise the longer chunk simply for having a larger-norm embedding.

HNSW Index

Comparing the query vector against all 6112 stored embeddings by brute force on every request would be acceptable at this scale, but ChromaDB instead uses the Hierarchical Navigable Small World (HNSW) graph index [5] to perform approximate nearest-neighbour search in sub-linear time. HNSW builds a multi-layer proximity graph over the stored vectors and navigates it greedily from a random entry point, converging on the nearest neighbours without examining every element. The result is a small, configurable approximation error in exchange for a large reduction in search latency as the collection grows.

```
1 _client      = chromadb.PersistentClient(path=CHROMA_PATH)
2 _collection = _client.get_collection(name=COLLECTION_NAME)
3 _model      = SentenceTransformer(EMBEDDING_MODEL)
4
5 def retrieve(query: str, n_results: int = TOP_K) -> list[dict]:
6     query_emb = _model.encode([query]).tolist()
7     results   = _collection.query(
8         query_embeddings=query_emb, n_results=n_results
9     )
10    return [
11        {"content": doc, "metadata": meta}
12        for doc, meta in zip(
13            results["documents"][0], results["metadatas"][0]
14        )
15    ]
```

Code 4.7: Retrieval function (app/rag.py)

The ChromaDB client and the SentenceTransformer model are initialised eagerly at module import time as module-level singletons. This means the model is loaded from disk once when the FastAPI process starts, and every subsequent request reuses the already-loaded objects, keeping per-query latency low.

Prompt Construction

The retrieved chunks are assembled into a structured prompt that instructs the LLM to answer only from the provided context and to admit when the answer is not available. Each chunk is labelled with its source file name so the model can associate claims with documents.

```
1 def build_prompt(query: str, chunks: list[dict]) -> str:
2     context = "\n\n".join(
3         f"[Source: {c['metadata']['file_name']}] \n{c['content']}"
4         for c in chunks
5     )
6     return (
7         "You are a helpful assistant for ELTE Faculty of
8         Informatics "
9         "students. Answer the question using only the context below
10        . "
11        "If the answer is not in the context, say so.\n\n"
12        f"Context:\n{context}\n\n"
13        f"Question: {query}\nAnswer:"
14    )
```

Code 4.8: Prompt template (`app/rag.py`)

The explicit instruction *“If the answer is not in the context, say so”* is important for a faculty assistant: it is better to acknowledge a gap in the indexed documents than to hallucinate a regulation or deadline.

Local LLM via Ollama

Answers are generated by a locally running language model served through Ollama. The default model is `llama3.2:3b` (3 billion parameters), but the system supports runtime model switching: the frontend exposes a model selector that queries the `GET /models` endpoint for the list of models currently available in the local Ollama installation, and the selected model name is passed with each chat re-

quest. During evaluation an additional model, `gemma3:4b`, was also tested. Ollama exposes a simple HTTP API, meaning the production code does not depend on any cloud service or API key. The model runs on the same machine as the application, which preserves user privacy and eliminates per-query costs.

```
1 def call_ollama(prompt: str, model: str = OLLAMA_MODEL) -> str:
2     if not check_ollama():
3         raise RuntimeError(
4             "Ollama is not running. Start it with: ollama serve"
5         )
6     payload = {
7         "model": model,
8         "prompt": prompt,
9         "stream": False,
10        "options": {"temperature": TEMPERATURE, "num_ctx": 2048},
11    }
12    try:
13        r = requests.post(OLLAMA_URL, json=payload, timeout=
14            TIMEOUT_S)
15        r.raise_for_status()
16        return r.json()["response"].strip()
17    except requests.exceptions.Timeout:
18        raise RuntimeError(
19            f"Ollama timed out after {TIMEOUT_S}s"
20        )
21    except requests.exceptions.HTTPError as e:
22        raise RuntimeError(
23            f"Ollama HTTP {e.response.status_code} -- "
24            f"is model '{model}' pulled?"
25        )
26    except KeyError:
27        raise RuntimeError(
28            f"Unexpected Ollama response: {r.text[:200]}"
29        )
```

Code 4.9: Ollama invocation (`app/rag.py`)

The context window is set to 2048 tokens, which comfortably fits three retrieved chunks plus the prompt template and leaves room for the generated answer. A `check_ollama()` health probe is called before every generation request. The `try/except` block distinguishes three failure modes: a timeout (model still

loading), an HTTP error code (model not pulled), and a malformed response body — each producing a descriptive `RuntimeError` that the API layer converts into a 503 response.

End-to-End Query Function

The three steps are composed into a single `rag_query` function that is called by the API layer:

```
1 def rag_query(query: str) -> dict:
2     chunks = retrieve(query)
3     prompt = build_prompt(query, chunks)
4     answer = call_ollama(prompt)
5     return {
6         "answer": answer,
7         "sources": [
8             {
9                 "chunk_id": c["metadata"]["chunk_id"],
10                "file": c["metadata"]["file_name"],
11            }
12            for c in chunks
13        ],
14    }
```

Code 4.10: End-to-end RAG query (`app/rag.py`)

The function returns both the answer text and a list of source references. This allows the frontend to display which documents were used, giving the user a way to verify the answer or read further.

4.7.6 FastAPI Backend

The backend is a FastAPI application (`app/main.py`) that exposes three HTTP endpoints and serves the static frontend files.

Method	Path	Description
GET	/	Serves the single-page chat interface (<code>index.html</code>).
GET	/health	Returns application status and whether Ollama is reachable.
GET	/models	Returns the list of models currently available in the local Ollama installation.
POST	/chat	Accepts a user message and optional <code>session_id</code> and <code>model</code> , runs the RAG pipeline, and returns the answer with source references.
GET	/sessions	Returns the list of past chat sessions ordered by most recently updated.
GET	/sessions/{id}/messages	Returns all messages in a specific session in chronological order.
DELETE	/sessions/{id}	Deletes a session and all its associated messages from the log.
POST	/upload	Accepts a user-uploaded file, chunks and embeds it, and adds it to the live ChromaDB collection.

Table 4.5: API endpoints exposed by the FastAPI backend.

The `/chat` endpoint wraps `rag_query` with error handling.

```

1 @app.post("/chat", response_model=ChatResponse)
2 def chat(req: ChatRequest):
3     try:
4         result = rag_query(req.message)
5     except RuntimeError as e:
6         raise HTTPException(status_code=503, detail=str(e))
7     return result

```

Code 4.11: Chat endpoint (`app/main.py`)

If `rag_query` raises a `RuntimeError` (for example because Ollama is not running or timed out), the endpoint converts it into an HTTP 503 response so the frontend can display a meaningful error rather than a timeout spinner. CORS middleware is enabled to allow the frontend to communicate with the backend. For a locally deployed application this is straightforward, but in a production web deployment the allowed origins would need to be restricted to a specific domain.

4.7.7 Logging System

Every chat interaction is recorded in a SQLite database at `data/logs/chat_logs.db`. The schema captures the timestamp, the user's message, the generated answer, the source references (as a JSON array), the response time in milliseconds, and any error message.

```
1 con.execute("""
2     CREATE TABLE IF NOT EXISTS chat_logs (
3         id            INTEGER PRIMARY KEY AUTOINCREMENT,
4         timestamp     TEXT      NOT NULL,
5         user_message  TEXT      NOT NULL,
6         answer        TEXT      NOT NULL,
7         sources       TEXT      NOT NULL, -- JSON array
8         response_ms   INTEGER NOT NULL,
9         error         TEXT      DEFAULT NULL,
10        session_id    TEXT      DEFAULT NULL
11    )
12 """)
13 con.execute("""
14     CREATE TABLE IF NOT EXISTS sessions (
15         session_id    TEXT PRIMARY KEY,
16         title         TEXT NOT NULL,
17         created_at    TEXT NOT NULL,
18         updated_at    TEXT NOT NULL
19     )
20 """)
21 con.execute("""
22     CREATE TABLE IF NOT EXISTS uploads (
23         id            INTEGER PRIMARY KEY AUTOINCREMENT,
24         timestamp     TEXT      NOT NULL,
25         file_name     TEXT      NOT NULL,
26         sha256        TEXT      NOT NULL,
27         size_bytes    INTEGER NOT NULL,
28         chunk_count   INTEGER NOT NULL,
29         status        TEXT      NOT NULL
30     )
31 """)
```

Code 4.12: Database schema initialisation (`app/logger.py`)

The `chat_logs` table links each message to a `session_id`, enabling per-session history retrieval. The `sessions` table stores one row per conversation — its title (the first 60 characters of the opening message) and timestamps — so the sidebar can list sessions without scanning the full message log. The `uploads` table records every user-uploaded file alongside its SHA-256 digest and the number of chunks produced, providing an audit trail for the live document ingestion feature.

4.7.8 Frontend

The user interface is a single-page HTML application served as a static file by FastAPI. It uses vanilla JavaScript without any framework dependency, which keeps the bundle small and removes the need for a separate build step.

The visual design follows the ELTE colour palette: navy blue (`#012850`) for the header, teal (`#009FA3`) for interactive elements, and light grey for the chat background. The layout consists of a fixed header bar, a scrollable message area, and a pinned input row at the bottom.

When the user sends a message, the frontend posts a JSON request to `/chat` and appends a typing indicator to the message area while waiting for the response. On success, both the answer text and the source file names are displayed in the assistant's reply bubble. On failure (HTTP 503), an error message is shown inline without leaving the page.

A status indicator in the header reflects the current Ollama availability by polling the `/health` endpoint on page load. This gives the user immediate feedback if the local model is not running before they attempt their first query.

4.7.9 Project Structure

The repository is organised into clearly separated concerns, as summarised in Table 4.6.

Path	Purpose
scripts/crawler.py	Resumable BFS web crawler for both ELTE domains.
scripts/chat_cli.py	Interactive terminal client for the chat API.
app/main.py	FastAPI application and all HTTP endpoints.
app/rag.py	Retrieval, prompt building, and Ollama integration.
app/ingest.py	File loading, chunking, and live embedding for user-uploaded documents.
app/logger.py	SQLite-backed chat and upload logging.
app/settings.py	Centralised configuration constants.
app/static/	Single-page HTML/CSS/JS frontend.
tests/	End-to-end pytest suite covering all API endpoints.
notebooks/01_data_ingestion.ipynb	Document loading, chunking, and manifest update.
notebooks/02_embedding.ipynb	Incremental embedding generation and ChromaDB update.
notebooks/03_RAG_pipeline.ipynb	Interactive exploration of the full RAG pipeline.
notebooks/04_evaluation.ipynb	Evaluation of retrieval and answer quality across four configurations.
data/raw/	Source documents (HTML; PDFs in <code>linked_pdfs/</code> ; DOCX in <code>linked_docx/</code>).
data/processed/	<code>chunks.json</code> , <code>manifest.json</code> , and ChromaDB vector store.
data/logs/	SQLite chat log database.
data/uploads/	User-uploaded documents stored after ingestion.

Table 4.6: Project directory structure and file responsibilities.

Chapter 5

Testing

This chapter covers testing and evaluation of the ELTE IK Assistant. It begins with functional testing of the API endpoints and pipeline, then presents an empirical system evaluation across four model and retrieval configurations, and concludes with a user evaluation section.

5.1 Functional Testing

The project includes an end-to-end test suite located in the `tests/` directory. Tests are written using `pytest` and interact with the live system over HTTP — there are no mocks — so they exercise the full request path from the API layer through ChromaDB retrieval to Ollama inference.

Test Architecture

All tests are tagged with the `@pytest.mark.e2e` marker, which keeps them separate from any future unit tests and allows selective execution. A session-scoped `warm_up_model` fixture fires one dummy `/chat` request before the first test runs, loading the LLM into RAM so that individual tests do not time out during model initialisation. Two further session-scoped fixtures — `live_server` and `ollama_up` — check the `/health` endpoint and skip the entire session with an actionable message if the server or Ollama is not reachable.

Coverage

Table 5.1 summarises the endpoints and behaviours verified by the suite.

Endpoint	Test	Assertion	Result
GET /health	server up	<code>status == "ok"</code> , <code>ollama</code> key present	Pass
GET /health	Ollama online	<code>ollama == true</code>	Pass
POST /chat	basic response	200, non-empty answer, sources list, <code>response_ms > 0</code>	Pass
POST /chat	in-scope query	answer length > 20 characters	Pass
POST /chat	out-of-scope query	200, non-empty answer	Pass
POST /chat	source keys	each source contains <code>chunk_id</code> and <code>file</code>	Pass
POST /chat	session logging	session appears in GET /sessions after chat	Pass
GET /sessions	session created	session ID present after chat with <code>session_id</code>	Pass
GET /sessions/{id}/messages	messages retrievable	one message with correct <code>user_message</code> field	Pass
DELETE /sessions/{id}	session delete	204, session absent from list afterwards	Pass
POST /upload	valid PDF	200, <code>status</code> is "indexed" or "already_indexed"	Pass
POST /upload	duplicate file	<code>status == "already_indexed"</code> on second upload	Pass
POST /upload	invalid extension	415 Unsupported Media Type	Pass
GET /info	collection size	<code>total_chunks</code> is a non-negative integer	Pass
GET /info	embedding model	<code>embedding_model == "all-MiniLM-L6-v2"</code>	Pass
GET /models	model list	<code>models</code> is a non-empty list, <code>default</code> key present	Pass

Table 5.1: End-to-end test cases and their assertions.

Because LLM output is non-deterministic, no test asserts exact answer text; all assertions target response structure, status codes, and field types. All 16 test cases passed on the reference hardware configuration running `llama3.2:3b` via Ollama.

Running the Tests

Before running the suite the following prerequisites must be met: Ollama must be running (`ollama serve`), the target model must be pulled (`ollama pull llama3.2:3b`), and the FastAPI server must be started (`uvicorn app.main:app --reload`). The suite is then executed with:

```
1 python -m pytest tests
```

Code 5.1: Database schema initialisation (`app/logger.py`)

If the server or Ollama is not available, all tests are automatically skipped rather than failed, and the skip message states the exact command needed to resolve the issue.

5.2 User Evaluation

5.3 System Evaluation

The following sections present an empirical evaluation of the system across four configurations, measuring answer quality, faithfulness to retrieved source material, out-of-scope refusal behaviour, retrieval effectiveness, and response latency. The goal is to quantify how much value the RAG component adds over a plain language model baseline, and to compare the two evaluated local models under both conditions.

5.3.1 Evaluation Setup

Configurations

Four configurations were evaluated in a factorial design crossing two models with two retrieval modes:

ID	Model	RAG	Description
A1	llama3.2:3b	No	Baseline: model knowledge only
B1	llama3.2:3b	Yes	RAG with top-3 chunk retrieval
A2	gemma3:4b	No	Baseline: model knowledge only
B2	gemma3:4b	Yes	RAG with top-3 chunk retrieval

Table 5.2: The four evaluation configurations.

Both models were run locally via Ollama with 4-bit quantisation and temperature 0.1. No internet access was used during evaluation.

Test Set

The test set consists of 20 questions divided into two categories:

- **15 in-scope questions** about topics present in the crawled ELTE documents: course prerequisite rules (strong and weak), Neptun registration behaviour, faculty staff information, the Computer Science BSc programme, and international partnerships. These questions have a definite correct answer that can be verified against the source documents.
- **5 out-of-scope questions** with no connection to the faculty corpus: general knowledge (*Who won the FIFA World Cup in 2022?*), weather, cooking, geography, and humour. These questions measure whether the system correctly declines to answer rather than hallucinating.

Reference answers for in-scope questions were written manually by consulting the source documents. Each question was run through all four configurations independently; the model received no memory of previous questions.

Metrics

ROUGE-L The longest-common-subsequence F1 score between the generated answer and the reference answer [6]. Captures lexical overlap with the ground truth on a scale from 0 to 1.

Faithfulness The proportion of words in the generated answer that appear in the retrieved chunks. A high faithfulness score indicates the model is drawing on the provided context rather than confabulating.

Refusal rate For out-of-scope questions only: the fraction of queries for which the model produced a non-answer (e.g. *I don't know* or *This is outside my knowledge*). Higher is better for this category.

Retrieval hit-rate For RAG configurations: the fraction of in-scope questions for which at least one of the retrieved chunks came from the expected source document. Measures the quality of the embedding-based retrieval independently of the language model.

Response time Wall-clock latency from request to complete response, measured in milliseconds on a laptop with 8 GB RAM and no GPU.

5.3.2 Results

Overall Metric Summary

Table 5.3 reports the per-configuration averages across all 20 test questions.

ID	Model	ROUGE-L	Faithfulness	Refusal	Hit-rate	Avg time
A1	llama3.2:3b, no RAG	0.117	0.377	0.00	0.667	47.6 s
B1	llama3.2:3b, RAG	0.436	0.711	0.60	0.667	70.5 s
A2	gemma3:4b, no RAG	0.078	0.355	0.00	0.667	121.1 s
B2	gemma3:4b, RAG	0.534	0.844	0.80	0.667	61.4 s

Table 5.3: Evaluation results averaged over 20 test questions. Hit-rate is computed over in-scope questions only (15 questions); refusal rate is computed over out-of-scope questions only (5 questions).

Effect of RAG on Answer Quality

Adding RAG context produces a consistent and substantial improvement in both lexical quality and faithfulness for both models.

- **ROUGE-L** increases by a factor of $3.7\times$ for llama3.2:3b ($0.117 \rightarrow 0.436$) and by $6.9\times$ for gemma3:4b ($0.078 \rightarrow 0.534$). The larger relative gain for Gemma suggests that without retrieval it relies more heavily on parametric knowledge, which is less aligned with ELTE-specific terminology.
- **Faithfulness** roughly doubles in both cases ($0.377 \rightarrow 0.711$ for Llama; $0.355 \rightarrow 0.844$ for Gemma), confirming that the retrieved chunks are genuinely shaping the generated text rather than being ignored.

The qualitative difference is visible in individual answers. For the question “*What is a strong prerequisite?*”, the no-RAG baseline (A1) produces a generic university definition with invented ELTE-specific examples that do not appear in any source document. The RAG-augmented configuration (B1) produces an answer drawn directly from the prerequisites document: “*A strong prerequisite is a prerequisite without which the follow-up subject cannot be taken in the next semester; the follow-up subject can only be registered for after the prerequisite subject has been fulfilled.*” The ROUGE-L score for this question rises from 0.13 (A1) to 0.29 (B1) and faithfulness from 0.44 to 0.80.

Model Comparison

Among the RAG configurations, **gemma3:4b** (B2) outperforms **llama3.2:3b** (B1) on both answer quality metrics: ROUGE-L 0.534 vs. 0.436, faithfulness 0.844 vs. 0.711. Gemma produces more concise answers that stay closer to the retrieved context, whereas Llama tends to expand its answers with surrounding commentary.

Among the no-RAG baselines, both models score similarly poorly, with Llama slightly ahead on ROUGE-L (0.117 vs. 0.078). Neither baseline reliably answers ELTE-specific questions correctly without retrieval.

Retrieval Quality

The retrieval hit-rate is 0.667 (10 out of 15 in-scope questions) and is identical across all four configurations, since the retrieval step is determined solely by the embedding model and vector index. The five missed questions share a common cause: they ask about topics (Erasmus partnerships, CEEPUS network, Dean contact details) that are present in the HTML pages but were truncated or de-prioritised during chunking because the relevant content appeared in shallow navigation-style text rather than in a dedicated content section. This is the primary bottleneck limiting further ROUGE-L and faithfulness gains: when the correct chunk is not retrieved, the language model must fall back on parametric knowledge, which tends to be inaccurate for institution-specific facts.

Out-of-Scope Refusal

Neither baseline model refuses any out-of-scope question: both A1 and A2 answer all five irrelevant queries confidently, generating plausible but entirely off-topic responses. Among the RAG configurations, refusal rates are substantially higher based on manual review of the outputs. **llama3.2:3b** with RAG (B1) correctly refuses 3 out of 5 out-of-scope questions (60%), and **gemma3:4b** with RAG (B2) refuses 4 out of 5 (80%). In both cases the refusal is triggered by the prompt instruction to answer only from provided context: when no relevant chunks are retrieved, the model recognises the absence of grounding material and declines to answer rather than falling back on parametric knowledge.

RAG therefore has a clear secondary benefit beyond answer quality: it substantially improves the system’s ability to decline out-of-scope questions. However, nei-

ther RAG configuration reaches 100% refusal, and the no-RAG baselines offer no protection at all, confirming that the prompt-level mechanism alone is not fully reliable.

Response Latency

Configuration	Avg (s)	No-RAG overhead	Req. threshold
A1 — llama3.2:3b, no RAG	47.6	—	exceeded
B1 — llama3.2:3b, RAG	70.5	+22.9 s	exceeded
A2 — gemma3:4b, no RAG	121.1	—	exceeded
B2 — gemma3:4b, RAG	61.4	−59.7 s	exceeded

Table 5.4: Average response latency. The non-functional requirement NF-4 sets a target of 30 seconds.

All four configurations exceed the 30-second latency target defined in the requirements (NF-4). This is expected on CPU-only hardware with 4-bit quantised models: the bottleneck is autoregressive token generation, which scales linearly with the number of output tokens. Two observations are worth noting. First, RAG adds latency for llama3.2:3b (+23 s) but *reduces* it for gemma3:4b (−60 s): when grounded by retrieved context, Gemma produces shorter, more focused answers, whereas without context it generates lengthy elaborations. Second, llama3.2:3b is substantially faster than gemma3:4b under both conditions, making it the more practical choice where latency matters more than marginal quality gains.

5.3.3 Discussion

RAG Effectiveness

The evaluation results confirm that RAG is effective for this use case. Grounding the language model in retrieved faculty documents more than triples ROUGE-L scores and nearly doubles faithfulness, for both of the evaluated models. The improvements are consistent and large enough to be meaningful in practice, not merely statistical noise over a 20-question benchmark.

The fundamental mechanism is straightforward: without context, neither llama3.2:3b nor gemma3:4b has reliable knowledge of ELTE-specific regulations — they were trained on general web text and have no exposure to the faculty’s prerequisite rules or administrative procedures. The retrieval step injects the relevant

passage directly into the prompt, allowing the model to produce a grounded answer even though it has no parametric knowledge of the topic.

Limitations

Several limitations should be acknowledged.

Small benchmark. Twenty questions is a limited evaluation set. The in-scope questions are skewed toward prerequisite rules because that is the richest and most structured document in the corpus. Topics such as scholarship regulations, exam appeal procedures, and credit transfer rules are underrepresented.

Retrieval ceiling. The 0.667 hit-rate means one third of in-scope questions are answered without relevant context. Improving chunking strategy (e.g. heading-aware splits, larger overlap) or adding a re-ranking step could raise this ceiling and correspondingly improve ROUGE-L and faithfulness.

ROUGE-L as a proxy. ROUGE-L measures lexical overlap with a manually written reference answer. A response that is semantically correct but uses different phrasing will score poorly. For a more robust evaluation a human-judged correctness rating or a semantic similarity metric (e.g. BERTScore) would complement ROUGE-L.

Latency. All configurations exceed the 30-second target. Achieving this target on CPU-only hardware with the current models would require either a smaller model (1B parameters) or a faster inference backend. For a deployment scenario where the server runs on shared faculty hardware with a GPU, latency would drop significantly.

Recommended Configuration

Based on the evaluation results, **gemma3:4b with RAG (B2)** is the recommended production configuration: it achieves the highest answer quality (ROUGE-L 0.534, faithfulness 0.844) and, counterintuitively, lower latency than the no-RAG Gemma baseline (61 s vs. 121 s). If latency is the primary constraint, **llama3.2:3b with RAG (B1)** offers a reasonable quality-speed trade-off at 70 seconds average response time with only a modest quality reduction.

Chapter 6

Conclusion

This thesis has presented the design, implementation, and empirical evaluation of a locally deployed retrieval-augmented generation chatbot for the ELTE Faculty of Informatics. The system allows students and staff to ask natural-language questions about the curriculum, course prerequisites, exam rules, and Neptun registration procedures, and receive answers grounded in official faculty documents — without sending any data to an external server.

6.1 Summary of Contributions

The thesis has delivered the following concrete artefacts and findings.

Document collection and ingestion pipeline. A resumable breadth-first crawler collects HTML pages, PDF documents, and DOCX files from the faculty’s public web presence. An incremental ingestion pipeline maintains a SHA-256 manifest so that only new or changed files need to be reprocessed on subsequent runs, reducing re-indexing cost to a fraction of a full rebuild. The resulting corpus of 482 source files is chunked into 6112 overlapping text segments and stored in a ChromaDB vector index.

Offline RAG pipeline. Each chunk is encoded into a 384-dimensional dense vector by the `all-MiniLM-L6-v2` sentence embedding model. At query time the top-3 most similar chunks are retrieved by cosine similarity and assembled into a structured prompt that instructs the language model to answer only from the

provided context. The entire pipeline runs on a standard laptop with 8 GB of RAM and no GPU.

Local language model serving. Two quantised local models — `llama3.2:3b` and `gemma3:4b` — are served by Ollama. No query, no retrieved document chunk, and no generated answer leaves the host machine at any point during inference, satisfying the privacy and offline-operation requirements that motivated the project.

Production-quality backend and frontend. A FastAPI backend exposes the RAG pipeline through a minimal REST API, logs every interaction to a local SQLite database, and serves a browser-based chat interface styled to match the faculty’s visual identity. The frontend supports session history, model switching, session deletion, and per-answer source references.

Empirical evaluation. A 20-question benchmark — 15 in-scope faculty questions and 5 out-of-scope control questions — was run across all four configurations. The results establish that RAG is the decisive factor in answer quality: ROUGE-L increases by $3.7\times$ for `llama3.2:3b` and by $6.9\times$ for `gemma3:4b` when retrieval context is added. Source faithfulness nearly doubles in both cases. The best-performing configuration, `gemma3:4b` with RAG, achieves ROUGE-L 0.534 and faithfulness 0.844, and produces shorter, more focused answers that also reduce average response latency relative to the no-RAG Gemma baseline.

6.2 Reflection on Goals

The system meets the two primary goals stated in Chapter 1.

Privacy. All computation — embedding, retrieval, and generation — occurs on the local machine. There is no network dependency after the initial model download, and no user query reaches any external service. This satisfies the GDPR-motivated offline-operation requirement.

Domain grounding. Without RAG, neither evaluated model reliably answers ELTE-specific questions: they have no parametric knowledge of faculty prerequisite rules or Neptun registration procedures. With RAG, both models produce answers that are substantially more accurate and faithful to the source documents. The

system successfully bridges the gap between what the language models know in general and what the faculty documents say specifically.

The 30-second response latency target (NF-4) was not met: all four configurations exceed it on CPU-only hardware, with median times ranging from 47 to 121 seconds. This is a known consequence of running billion-parameter models on a laptop without GPU acceleration, and is noted as a deployment constraint rather than an architectural failure.

6.3 Limitations

Three limitations are worth restating clearly.

Retrieval ceiling. The embedding-based retrieval achieves a hit-rate of 0.667, meaning one in three in-scope questions is answered without the most relevant chunk. The missed questions concern topics whose source content appears in shallow navigation text that is not well-represented after chunking. Improving chunking strategy or adding a re-ranking step is the highest-leverage improvement available.

Out-of-scope refusal. Neither baseline model refuses any out-of-scope question; both answer all five irrelevant queries confidently without retrieval context. Among the RAG configurations, manual review of outputs shows that `llama3.2:3b` with RAG correctly refuses 3 out of 5 out-of-scope questions (60%) and `gemma3:4b` with RAG refuses 4 out of 5 (80%). The prompt-level instruction to answer only from provided context is therefore the primary driver of refusal behaviour — but it is not fully reliable. A more robust mechanism — such as a retrieval-score threshold below which the system returns a fixed “I don’t know” response — is needed for a production deployment.

Benchmark scale. The 20-question test set is small and skewed toward pre-requisite rules. A broader evaluation covering scholarships, exam appeals, credit transfer, and admissions would give a more complete picture of system performance.

6.4 Future Work

Several directions could extend the system beyond its current scope.

Improved retrieval. Replacing the fixed top- k retrieval with a cross-encoder reranker would improve chunk selection precision without changing the embedding index. Hybrid retrieval — combining dense vector search with a sparse BM25 index — is a well-established technique for handling exact-match queries (course codes, regulation numbers) that embedding models handle less reliably than semantic queries.

Retrieval-score threshold for refusal. A simple and robust out-of-scope detector would compare the cosine similarity of the top retrieved chunk against a calibrated threshold. If no chunk clears the threshold the system returns a fixed refusal response rather than forwarding a contentless prompt to the language model. This would eliminate the model-dependent refusal inconsistency observed in the evaluation.

Larger model on shared hardware. The latency constraint is a hardware constraint, not a software one. If the system were deployed on a faculty server with a mid-range GPU (e.g. NVIDIA RTX 3060, 12 GB VRAM), a 7B-parameter model could run at sub-10-second latency, bringing the system well within the original 30-second target while further improving answer quality.

Multilingual support. The faculty serves students who write questions in Hungarian as well as English. Replacing `all-MiniLM-L6-v2` with a multilingual sentence encoder (e.g. `paraphrase-multilingual-MiniLM-L12-v2`) and a multilingual instruction-tuned model would extend the system to Hungarian-language queries without re-indexing the corpus.

User feedback loop. The SQLite interaction log provides a natural foundation for a feedback mechanism: users could mark an answer as helpful or unhelpful. Aggregated feedback could be used to identify retrieval failures, surface low-quality chunks for manual review, and prioritise document updates in the corpus.

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